

Lecture(11)

Chapter(3)

Function

- Identity function
- Inverse functions
- Compositions of functions

Type of functions

Identity function

Definition: Given a set A , the identity function on A is the function

$i_A : A \rightarrow A$ defined as $i_A(x) = x$ for every $x \in A$.

Example: If $A = \{1, 2, 3\}$, then $i_A = \{(1, 1), (2, 2), (3, 3)\}$. Also $i_{\mathbb{Z}} = \{(n, n) : n \in \mathbb{Z}\}$. The identity function on a set is the function that sends any element of the set to itself.

Notice that for any set A , the identity function i_A is bijective: It is injective because $i_A(x) = i_A(y)$ immediately reduces to $x = y$. It is surjective because if we take any element b in the codomain A , then b is also in the domain A , and $i_A(b) = b$.

Type of functions

Inverse function

Now consider a one-to-one correspondence f from the set A to the set B . Because f is an onto function, every element of B is the image of some element in A . Furthermore, because f is also a one-to-one function, every element of B is the image of a *unique* element of A . Consequently, we can define a new function from B to A that reverses the correspondence given by f

Definition:

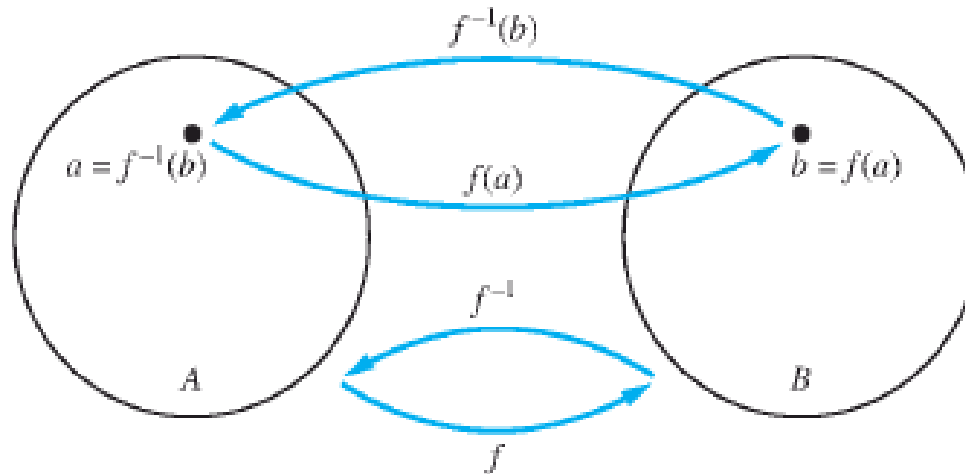
Let f be a one-to-one correspondence from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Remark: Be sure not to confuse the function f^{-1} with the function $1/f$, which is the function that assigns to each x in the domain the value $1/f(x)$. Notice that the latter makes sense only when $f(x)$ is a non-zero real number.

Type of functions

A bijective is called **invertible** because we can define an inverse of this function.

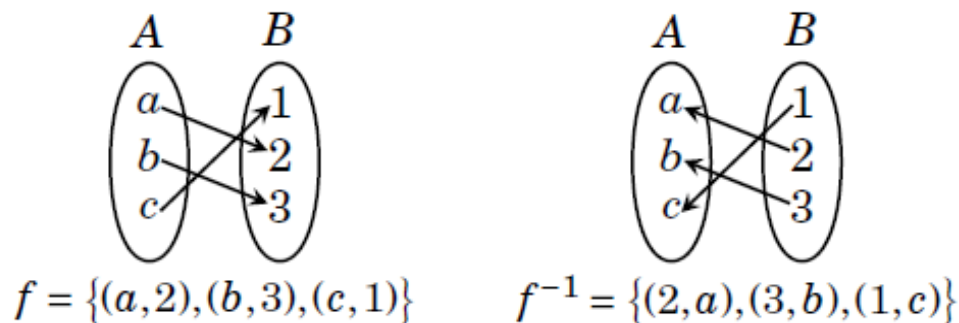
A function is **not invertible** if it is not bijective, because the inverse of such a function does not exist.



The Function f^{-1} Is the Inverse of Function f .

Type of functions

Example: let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$, and suppose f is the relation $f = \{(a, 2), (b, 3), (c, 1)\}$ from A to B . Then $f^{-1} = \{(2, a), (3, b), (1, c)\}$ and this is a relation from B to A . Notice that f is actually a function from A to B , and f^{-1} is a function from B to A . These two relations are drawn below. Notice the drawing for relation f^{-1} is just the drawing for f with arrows reversed.



Type of functions

Example:

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$. 

Example: Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution: Because $f(-2) = f(2) = 4$, f is not one-to-one. If an inverse function were defined, it would have to assign two elements to 4. Hence, f is not invertible. (Note we can also show that f is not invertible because it is not onto.) 

Type of functions

Example: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^3 + 1$ is bijective.

Find its inverse.

Solution:

We begin by writing $y = x^3 + 1$. Now interchange variables to obtain $x = y^3 + 1$. Solving for y produces $y = \sqrt[3]{x-1}$. Thus

$$f^{-1}(x) = \sqrt[3]{x-1}.$$

(You can check your answer by computing

$$f^{-1}(f(x)) = \sqrt[3]{f(x)-1} = \sqrt[3]{x^3+1-1} = x.$$

Therefore $f^{-1}(f(x)) = x$. Any answer other than x indicates a mistake.)

Type of functions

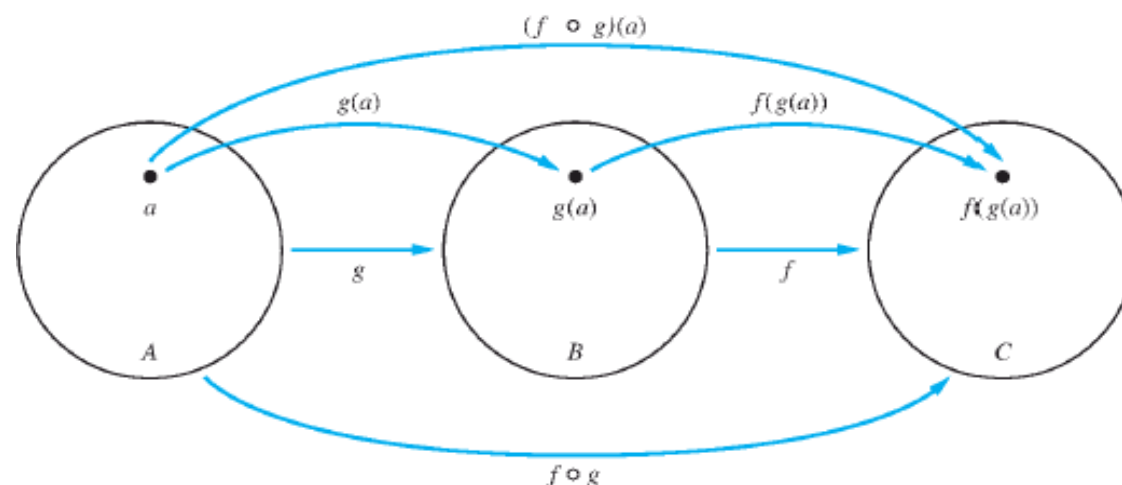
Composition of function

Definition:

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The *composition* of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$

In other words, $f \circ g$ is the function that assigns to the element a of A the element assigned by f to $g(a)$. That is, to find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$. Note that the composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .



Type of functions

Example:

Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution: The composition $f \circ g$ is defined by $(f \circ g)(a) = f(g(a)) = f(b) = 2$, $(f \circ g)(b) = f(g(b)) = f(c) = 1$, and $(f \circ g)(c) = f(g(c)) = f(a) = 3$.

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g . ◀

Example:

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution: Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11. \quad \blacktriangleleft$$

Recurrence relation

The simplest and most concrete type of recursive object in mathematics is a *recurrence relation*. Suppose we wish to define a function

$$P : \mathbf{N} \longrightarrow \mathbf{Z}$$

that inputs a natural number and returns an integer. The easiest way to do this is to give an explicit formula:

$$P(n) = \frac{n(n+1)}{2}.$$

To evaluate $P(n)$ for some given n , you just plug n into the formula.

It is always nice to have an explicit formula for a function, but sometimes these are hard to come by. Often a function comes up in mathematics that is natural to define recursively. Here is a second way of defining our function $P(n)$:

$$P(n) = \begin{cases} 1 & \text{if } n = 1 \\ n + P(n-1) & \text{if } n > 1. \end{cases} \quad \Rightarrow \quad \text{Equation 1}$$

167

This is a recursive definition because P is defined in terms of itself: P occurs in the formula that defines P

Recurrence relation

Example: Use Equation 1 to compute $P(5)$

Solution:

$$\begin{aligned}P(5) &= 5 + P(4) \\&= 5 + 4 + P(3) \\&= 5 + 4 + 3 + P(2) \\&= 5 + 4 + 3 + 2 + P(1) \\&= 5 + 4 + 3 + 2 + 1 \\&= 15.\end{aligned}$$

Exercises

1.

Consider the following recurrence relation:

$$H(n) = \begin{cases} 0 & \text{if } n \leq 0 \\ 1 & \text{if } n = 1 \text{ or } n = 2 \\ H(n-1) + H(n-2) - H(n-3) & \text{if } n > 2. \end{cases}$$

(a) Compute $H(n)$ for $n = 1, 2, \dots, 10$.

2.

Suppose $A = \{0, 1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$ and $f = \{(0, 3), (1, 3), (2, 4), (3, 2), (4, 2)\}$. State the domain and range of f . Find $f(2)$ and $f(1)$.

3.

There are four different functions $f : \{a, b\} \rightarrow \{0, 1\}$. List them all.

Exercises

4.

Give an example of a relation from $\{a, b, c, d\}$ to $\{d, e\}$ that is not a function.

5.

Suppose $A = \{1, 2, 3\}$. Let $f: A \rightarrow A$ be the function $f = \{(1, 2), (2, 2), (3, 1)\}$, and let $g: A \rightarrow A$ be the function $g = \{(1, 3), (2, 1), (3, 2)\}$. Find $g \circ f$ and $f \circ g$.

6. What is the inverse of the function $f(x) = 5x - 4$